

Engineering Economics

(MG3008)

Date: May 25th 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba

Final Exam

Total Time (Hrs): 3

Total Marks: 120

Total Questions: 5

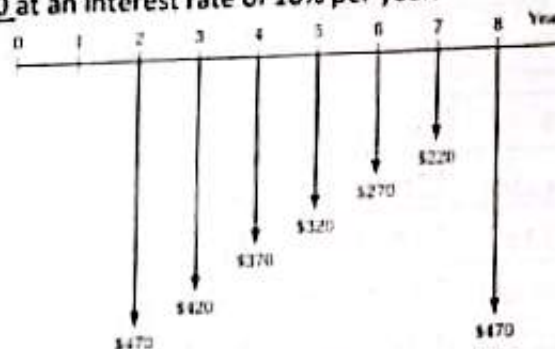
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Attempt all the questions. Show each step and all the calculations.

In those question where the table is provided fill the values in them and return the question paper with answer sheet.

CLO #: 2 Interpret the concept of time value of money

Q1: By applying time value of money for the cash flows shown in the diagram given below, determine the present worth in year 0 at an interest rate of 10% per year. [20 marks]



CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q2: A consulting engineer is analyzing two methods for lining ponds used for evaporating concentrate generated during reverse osmosis treatment of brackish ground water for the clay county industrial park. A geosynthetic bentonite clay liner, good for 3 years, will require a boiler costing \$920 plus valves costing \$360. The cost of water, treatment charges, electricity, etc will be \$3.1 per hour. Alternatively a high-density polyethylene geomembrane will cost \$3850 to buy, will have a useful life of 5 years, and will cost \$1.28 per hour to operate. What is the minimum number of hours per year that the cooling system must be used in order to justify purchase of the geosynthetic bentonite clay liner system? The MARR is 10% per year, and the salvage values are negligible. [20 marks]

CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q3: A construction company bought a 180,000 metric ton earth sifter at a cost of \$65,000. The company expects to keep the equipment a maximum of 5 years. The operating cost is expected to follow the series described by $40,000 + 10,000k$, where k is the number of years since it was purchased ($k=1,2,3,4,5$). The salvage value is estimated to be \$30,000 for years 1 and 2 and \$20,000 for years 3 through 5. At an interest rate of 10% per year, Compute the economic service life and the associated equivalent annual cost of the sifter. [20 marks]

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CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q4-a: Construct a loan repayment table for an asset worth \$1,500,000, interest is 14% per year for 6 years; when equal end of year payments are made. [15marks]

N	Big-Bal	Interest	Payment	Principal	Remaining	Acc- Principal
0					1,500,000	
1	1500.000	210000	385736	175736	1324264	175736
2	1324264	185396	385736	200339	1123925	376075
3	1123925	157349	385736	228386	895539	604461
4	895539	125375	385736	260360	635179	864821
5	635179	88925	385736	296810	338369	1161631
6	338369	47371	385736	338364	5	1499995

Q4-b: The Marginal tax rates are given, compose the remaining columns of the table: [5 Marks]

Range	Tax Rate	Limit-From	Limit-Up-to	Maximum Tax of Range	Maximum Tax Incurred
600,000	0%	0	600,000	0	0
600,000	1%	600,001	1,200,000	6000	6000
1,000,000	11%	1,200,001	2,200,000	110000	116000
1,000,000	23%	2,200,001	3,200,000	230000	346000
900,000	30%	3,200,001	4,100,000	270000	616000
Above	35%	4,100,001	Above	Above	Above.

CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q5-a: Create a MACRS table with 200% DB also calculate the depreciation amount and book value for each year of an asset, bought for \$1,500,000 and has a life of 5 years. [20 marks]

N	Bal	Dep1	Remain	N*	Straight	Dep	Currency	BV
1	100	20	80			20	300,000	1200000
2	80	32	48	4.5	17.77	32	480,000	720000
3	48	19.2	28.8	3.5	13.71	19.2	288,000	432000
4	28.8	11.52	17.28	2.5	11.52	11.52	172800	259200
⑤	17.28	6.912	10.37	1.5	11.52	11.52	172800	86400
6	10.37	4.15	0	0.5	1.15	1.15	17250	69150

Q5-b: Carry Forward from Q4 and Q5(a): From information calculated in table for interest in Q4-a; also use the MACRS table of Q5-a, and make the rest of the income statement using the tax table in Q4-b, with Revenue, Material, Labor and Overhead and their relative inflations given below: [20marks]

	"q"	0	1	2	3	4	5
Revenue	15	2,500,000	2875000	3306250	3802187	4372515	5028392
Material	9	200,000	218000	237620	259005	282316	307724
Labor	8	300,000	324000	349920	377913	408146	440797
Overhead	8	100,000	108000	116640	125971	136048	146932
RMC			2225000	2602070	3039298	3546005	4132939
Dep			300,000	480,000	288,000	172800	86400
EBIT			1925000	2122070	2751298	3373205	4046539
Interest			210000	185396	157349	125375	88925
EBT			1715000	1936674	2593949	3247830	3957614
Tax			62650	87034	206608	360349	573284
EAT			1652350	1849640	2387341	2887481	3384330

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$$100 + 15 = \frac{115}{100} = 1.15$$

$$\frac{109}{100} = 1.09$$

$$\frac{108}{100} = 1.08$$